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Interrogating the structure and function of the human voltage-gated proton channel (hH_v1) with a fluorescent noncanonical amino acid

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eLife Assessment

This elegant study presents a **valuable** approach to probing the structural features of the full-length human Hv1 channel as a purified protein, supported by rigorous biochemical assays and spectral FRET analysis, which will interest biophysicists and physiologists studying Hv1 and other ion channels. Overall, the work introduces an interesting labeling strategy and provides methodological observations that are of value in investigating hHv1. However, the analysis appears **incomplete**, requiring additional structural interpretation and mechanistic insight.

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Abstract

The human voltage-gated proton channel (hH_v1) is a dimer of voltage-sensor domains (VSDs) containing highly selective proton permeation pathways in each monomer. In addition to voltage, hH_v1 is regulated by other stimuli, including pH gradients, mechanical forces, and ligands such as Zn²⁺. Aside from the VSDs, this membrane protein contains an N-terminal domain and a C-terminal coiled-coil domain (CC) formed between the monomers. To address the need for direct measurements of conformational rearrangements in hH_v1, we developed a Förster resonance energy transfer (FRET) approach to measuring the conformational rearrangements in full-length hH_v1 purified from *E. coli*. We used genetic code expansion (GCE) to generate a library of 14 full-length hH_v1 constructs, each incorporating the fluorescent noncanonical amino acid acridon-2-ylalanine (Acd) at a different site throughout the various structural domains. Following the expression and purification of these hH_v1-Acd proteins, we found that 12 sites yielded stable and functional proton-permeable channels. The fluorescence properties of Acd at each site showed small site-specific differences. Furthermore, we measured site-specific FRET efficiencies from tryptophan (Trp) and tyrosine (Tyr) to Acd in the hH_v1-Acd proteins and found results consistent with correct folding in detergent micelles. Finally, the addition of Zn²⁺ produced reversible changes in FRET, with affected residues clustered on the intracellular side of the channel.

Introduction

The human voltage-gated proton channel (hH_v1) is a membrane protein that forms a proton selective channel (1, 2). Despite its physiological importance, its molecular mechanisms are still poorly understood, including the precise structural identity of the permeation pathway and the gating mechanism of voltage (3, 4), pH gradients (3, 5), membrane stretch (6), and ligands such as the classical inhibitor Zn²⁺ (7, 8). Understanding these processes requires connecting the functional properties to the hH_v1 structure.

Each monomer of hH_v1 dimers includes an intracellular N-terminal domain of unknown function, a transmembrane voltage-sensor domain (VSD), and an intracellular coiled-coil (CC), which forms the primary intersubunit interface (9, 10). Current structural models, however, have been obtained from truncated or chimeric channel constructs (11, 12), and thus, it is not clear whether they correspond to physiological or functional conformations. Moreover, the multiple kinetic components of H_v1 gating currents suggest that the protein's resting state comprises an ensemble of conformations (4, 5), which is consistent with the conformational flexibility inferred from structural approaches (12–14). Because protein function arises from this distribution of conformational states (15), approaches to directly measure structural heterogeneity in the full-length hH_v1 are needed.

An attractive approach to interrogate the structural heterogeneity of hH_v1 is fluorescence spectroscopy, which offers high sensitivity at nanomolar protein concentrations and compatibility with near-physiological conditions. Fluorescence can report on local environment and accessibility changes, as well as quantify distance changes through Förster resonance energy transfer (FRET) (16). Importantly, time-resolved FRET measured using lifetimes preserves information about conformational heterogeneity on the nanosecond timescale, which is lost in steady-state measurements due to the averaging process (17). Together, these capabilities make fluorescence, especially time-resolved FRET, a powerful method to connect hH_v1 structure and function (18).

A few obstacles remain to directly measuring conformational heterogeneity of hH_v1. First, a method to express and purify stable and functional protein is needed, which is challenging for membrane proteins. Second, site-specific fluorophore labeling is difficult at buried sites, resulting in low labeling efficiency. Third, traditional fluorophores are bulky and have long linkers (14), resulting in protein structure perturbation and heterogeneity dominated by the properties of the label (19). A recent method was optimized for robust expression and purification of full-length, functional hH_v1 in *E. coli* (20, 21). The remaining obstacles can be overcome by labeling the protein with the fluorescent noncanonical amino acid acridon-2-ylalanine (Acd) using genetic code expansion (GCE) (22–24). This technology allows site-specific labeling during translation by reassigning an amber stop codon using an orthogonal aminoacyl-tRNA synthetase (RS)/tRNA (25, 26) (Figure 1A). Acd's small size and minimal linker reduce perturbations and heterogeneity, and its favorable photophysics support both steady-state and time-resolved measurements (27). Here, we incorporated Acd across all hH_v1 structural domains and purified 12 of 14 constructs as stable, functional proteins. We then used Acd's environmental sensitivity and FRET from tryptophan (Trp)/tyrosine (Tyr) to Acd to validate the protein folding and report Zn²⁺-dependent conformational changes. These data establish a platform for interrogating conformational heterogeneity and dynamics in the full-length hH_v1.

Results

We generated a library of hH_v1 constructs with Acd incorporated across the different structural domains

Because available H_v1 structural models were produced from either truncated or modified proteins (11, 12), we used an AlphaFold model of the full-length, dimeric hH_v1 to guide Acd site selection (28, 29) (Figure 1B and Dataset S1). The model recapitulates the expected hH_v1 architecture with high confidence: S0 as a short helix parallel to the membrane, S1-S4 transmembrane segments forming a canonical VSD, and a C-terminal helix extending from S4 that forms the CC intersubunit interface. In addition, two helices in the N-terminal domain (P1 and P2) were predicted (Figure 1C), albeit with low confidence. The dimer interface in the membrane is primarily formed by S4-S4 contacts (Figure S1A), although S1-S1 contacts have been observed experimentally (9, 13). This model appears to represent an intermediate state, with the first S4 positive charged residue R208 proximal to D112 in the selectivity filter, and the third S4 positive

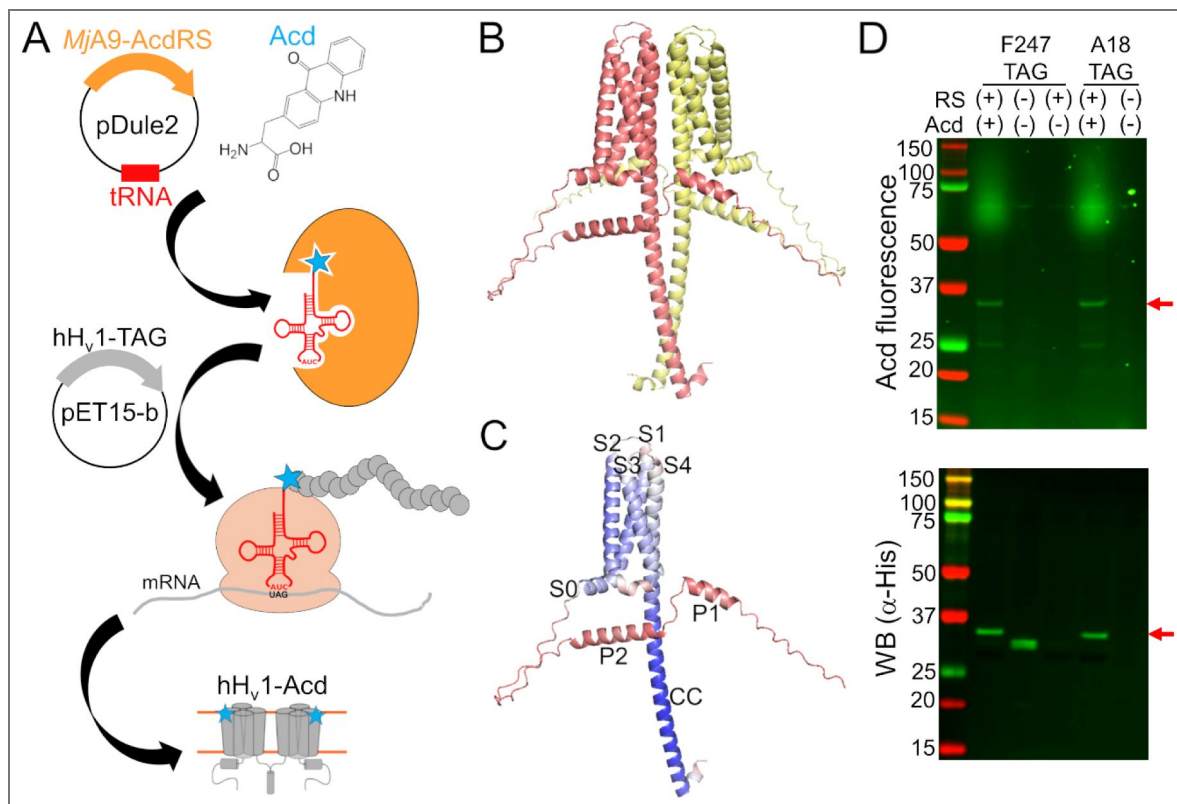


Figure 1. hHv1 was labeled with the fluorescent noncanonical amino acid Acrid by genetic code expansion.

(A) Cartoon showing the components of the genetic code expansion system used to incorporate Acrid via amber codon suppression in hHv1. (B) AlphaFold structural model of the dimer hHv1 colored by subunit. (C) AlphaFold structural model of a single hHv1 subunit colored by the predicted Local Distance Difference Test (pLDDT). Model confidence was represented by a gradient from red to blue, indicating low to high pLDDT. (D) In-gel fluorescence and Western blot to evaluate Acrid incorporation. An Acrid fluorescence band corresponding to the molecular weight of full-length hHv1 (red arrow) was observed in cellular extracts after separation by SDS-PAGE only when cells contained the *MjA9-AcdRS*/tRNA plasmid (RS) and were grown in the presence of 1mM Acrid (top). The Western blot against the His-tag present at the N-terminus of hHv1 confirmed that the band corresponded to the recombinant protein (bottom). Note the hHv1 truncated product produced when the amber stop codon is replacing F247 in the absence of the aminoacyl-tRNA synthetase/tRNA plasmid and Acrid.

charged residue R211 proximal to F150 in the charge-transfer center (Figure S1B [↗](#)). Although the model's accuracy and functional state are unknown, we used it as a rough starting point for selecting positions to incorporate Acd.

To test the feasibility and specificity of Acd incorporation in hH_v1, we initially selected two positions in the intracellular domains: A18 in the N-terminal domain and F247 in the CC. For each position, the wild-type codon was replaced with an amber stop codon (TAG). Bacteria co-transformed with the Acd RS/tRNA pair and hH_v1-TAG constructs expressed fluorescent full-length protein only when Acd was present in the culture medium (Figure 1D [↗](#)). We next generated a library of 14 constructs in which Acd was incorporated into every structural domain: the N-terminal domain, each helix of the VSD, and the CC (Figure 2A-B [↗](#)). Remarkably, most of the 14 hH_v1-Acd proteins expressed at levels comparable to the control (No TAG) (Figure 2C [↗](#) and S2 [↗](#)). As expected, several hH_v1-Acd proteins also produced TAG-truncated products (Figure 2C [↗](#)). Increasing the Acd concentration in the *E. coli* growth medium did not reduce the fraction of truncated protein (Figure S3 [↗](#)). One of the constructs, hH_v1-C107Acd, showed a weak in-gel fluorescence signal, although the Western blot signal was comparable to that of other constructs.

We successfully purified stable and functional hH_v1 labeled with Acd at 12 of the 14 selected positions

We attempted to purify all expressed hH_v1-Acd constructs by immobilized metal affinity chromatography using the detergent Anzergent 3-12 ([20](#), [21](#)). For 12 of 14 constructs, purification yielded a single fluorescent band on SDS-PAGE (Figure 3A [↗](#) and S4 [↗](#)). In contrast, no purified protein was detected for the V187Acd or Q233Acd constructs (Figure S4 [↗](#)). Because hH_v1-Q233Acd was robustly expressed (Figure 2C [↗](#)), its loss during purification suggests reduced stability. The hH_v1-Acd yields were construct-dependent (Figure 3B [↗](#)), and the TAG-truncated products did not co-purify with the full-length protein (Figure 3A [↗](#) and S4 [↗](#)).

We then evaluated the stability and homogeneity of the hH_v1-Acd proteins using fluorescence-detection size exclusion chromatography (FSEC) ([30](#)). Similar to the control (Figure S5 [↗](#)), most hH_v1-Acd proteins showed a main peak (~11.5 mL) followed by a minor peak at higher elution volumes (Figure 3C [↗](#) and S6 [↗](#); Table S1 [↗](#)). The proportion of these two peaks varied between constructs (Figure S6 [↗](#)). To determine whether the purified hH_v1-Acd proteins were functional, we reconstituted the FSEC fractions containing both peaks in asolectin liposomes and assessed their function using a liposome proton flux assay ([20](#), [31](#)). All 12 reconstituted constructs produced ACMA fluorescence quenching upon valinomycin addition, indicating that our purified proteins formed functional proton-permeable channels (Figure 3D [↗](#) and S7 [↗](#)). These results highlight the versatility of GCE for incorporating Acd as a fluorescent probe for structural studies of hH_v1.

Acd was relatively insensitive to the hydrophobicity of its local environment

To explore how the local environment can alter Acd's spectral properties, we measured the emission spectra of the free amino acid in different solvents (Figure 4A [↗](#) and Table S2 [↗](#)). The environmental sensitivity of Acd has been studied previously ([24](#), [27](#), [32](#)), but we expanded this work here by using the aprotic solvent ethyl acetate (EtAc) and a series of alcohols with varying alkyl chain lengths. Consistent with a general solvent effect produced by polarity ([16](#)), the emission spectrum was blue-shifted by approximately 20 nm in EtAc, the least polar solvent measured, relative to water (Figure 4A [↗](#) and Table S2 [↗](#)). Alcohols produced a consistently smaller blue shift, independent of their alkyl-chain length (Figure 4A [↗](#)). The spectrum in Buffer-H2, our standard experimental solution containing detergent micelles, was only very slightly blue-shifted relative to water (Figure 4A [↗](#)), suggesting that Acd did not substantially partition into the hydrophobic core of the detergent micelles. We also tested whether solvent polarity affects the fluorescence lifetime of Acd using time-correlated single photon counting (TCSPC). The

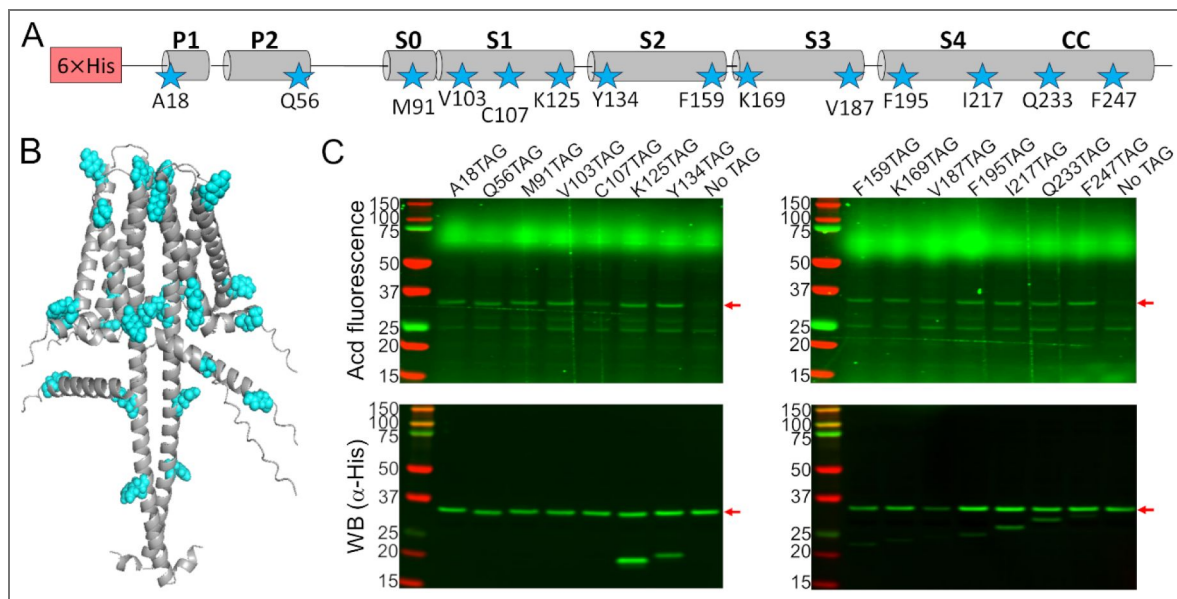


Figure 2. Acd was incorporated into 14 positions in the hHv₁ sequence.

(A) Cartoon showing the amino acids selected to be replaced by an amber stop codon in the hHv₁ secondary structure to incorporate Acd (stars). (B) AlphaFold dimer hHv₁ structural model with the amino acids selected to be replaced by Acd as cyan spheres. (C) Acd fluorescence gel (top) and Western blot (bottom) of cellular extracts after separation by SDS-PAGE showing the expression of the Acd-labeled hHv₁ mutants. The No TAG lane contains an extract from cells grown under identical conditions (in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair) and expressing hHv₁ without an amber stop codon.

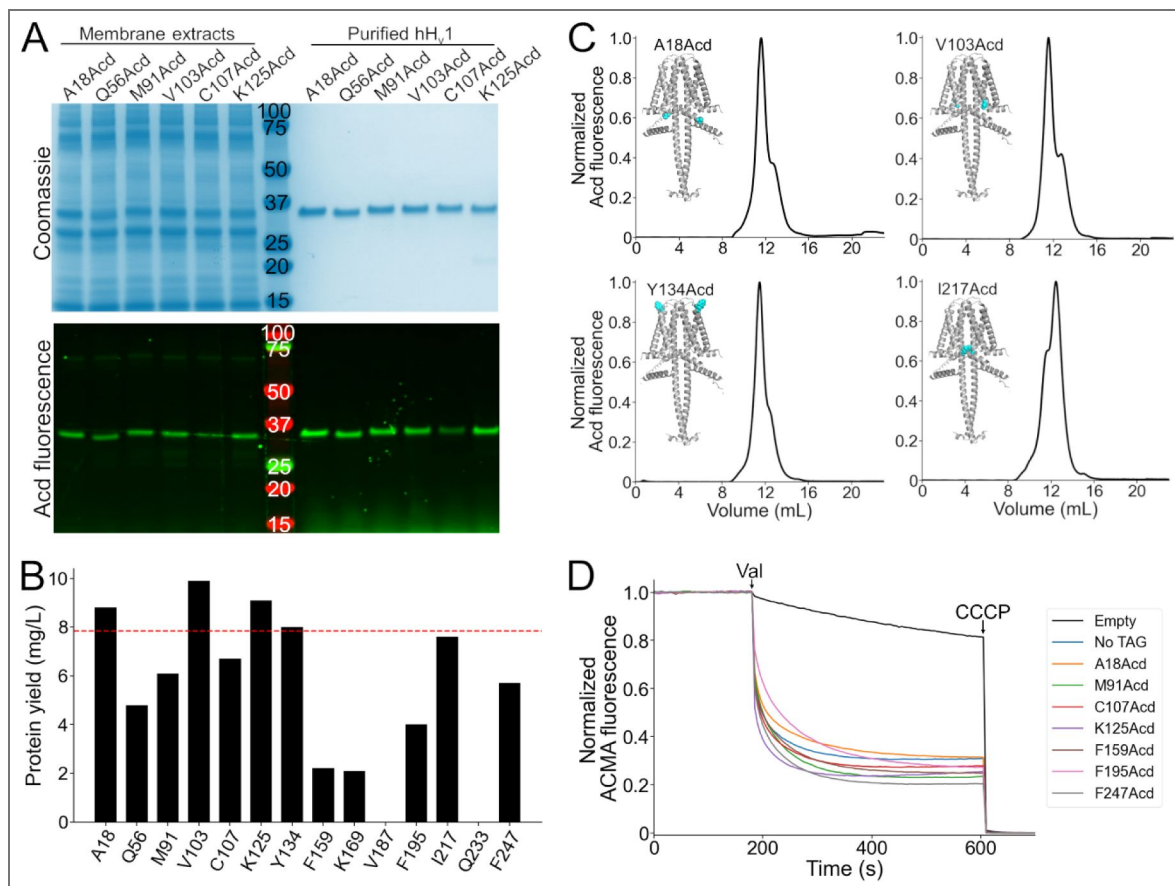


Figure 3. Purification and functional measurements of hHv1-Acd proteins.

(A) Coomassie-stained (top) and Acd fluorescence (bottom) gels after separation by SDS-PAGE showing representative samples of the membrane extracts after solubilization (left) and the final purified protein after immobilized metal affinity chromatography (right). (B) Protein yield of the hHv1 protein with Acd replacing the amino acid at the indicated position. Note that V187Acd and Q233Acd contained no detectable protein after purification. The red line indicates the protein yield obtained from hHv1 without an amber stop codon in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair (No TAG). (C) Representative fluorescence-detection size-exclusion chromatograms of the purified hHv1 proteins with Acd incorporated at the indicated amino acid position. Insets: AlphaFold dimer models, with the amino acid replaced by Acd shown as cyan spheres. (D) Representative liposome proton flux assay of asolectin proteoliposomes containing the indicated hHv1 protein. All purified proteins produced ACMA fluorescence quenching after the addition of valinomycin (Val). The protonophore CCCP was added at the end of the experiment as a control. The No TAG sample corresponds to proteoliposomes containing hHv1 without an amber stop codon expressed in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair.

fluorescence lifetime of Acd showed a similar trend to the spectral shifts (Figure 4B): EtAc shortened the lifetime markedly, Acd exhibited shorter lifetimes in alcohols than in water, and Acd in Buffer-H2 had a lifetime similar to that measured in water.

The spectral blue-shift and short lifetime of Acd in EtAc suggest that Acd may be able to report on the local environment at different sites in hH_v1, especially those that are surface-exposed versus those that face the lipid core. The emission spectra of the hH_v1-Acd proteins showed small, site-specific shifts (Figure 4C and Table S3). The local environment of Acd in hH_v1 spanned the range between those observed for free Acd in Buffer-H2 and in alcohols, with Q56Acd in the N-terminal intracellular domain and C107Acd in S1 showing the most red-shifted and blue-shifted spectra, respectively (Figure 4C). Consistent with the spectra, the fluorescence lifetimes of the hH_v1-Acd proteins fell between those measured for free Acd in Buffer-H2 and in alcohols, with Q56Acd exhibiting the slowest, and C107Acd the fastest, lifetimes (Figure 4D). While small, the differences in spectral properties of Acd in hH_v1 agree with their predicted location in the structural model, with C107 being the site closest to the detergent micelles' hydrophobic core. The low sensitivity to polarity changes of Acd in different hH_v1 local environments makes it well suited for measuring FRET (27, 33).

FRET between Trp/Tyr and Acd in hH_v1 suggested that the proteins are properly folded

Trp and Tyr have fluorescence emission spectra that overlap with the Acd absorption spectrum and therefore can act as FRET donors to Acd, with relatively short R_0 values (23.5 Å for Trp; 20.9 Å for Tyr) (22) (Figure 5A). Each hH_v1 monomer contains four Trp and four Tyr residues (yellow spheres in Figure 5B). In this structural model, the Trp residues are located in the N-terminal domain (W4, W38, and W45) and midway along transmembrane segment S4 in the VSD (W207). The Tyr residues are located in the N-terminal domain (Y35 and Y42) and near the extracellular end of transmembrane segment S2 (Y134 and Y141).

To quantify Trp/Tyr to Acd FRET, we used spectral FRET analysis (34). The total emission spectrum from Trp/Tyr and Acd was collected (Figure 5C). The Acd emission spectrum was extracted by subtracting a scaled Trp/Tyr spectrum collected from control wild-type protein ($F_{280, NoTAG}$) (Figure 5D and S8). For each Acd-containing construct, the ratio of the extracted spectrum ($F_{280} - F_{280, NoTAG}$) to the Acd spectrum with direct excitation (F_{370}) was calculated as *Ratio A* (Figure 5D and 5E). Because *Ratio A* is not wavelength-dependent, it reports the linearity of the detectors and the absence of significant contamination from other sources of fluorescence. The *Ratio A* component caused by the direct excitation of Acd (termed *Ratio A₀*) was measured with free Acd (35) (Figure 5E and S9). The difference between *Ratio A* and *Ratio A₀* is directly proportional to the FRET efficiency.

As expected, the *Ratio A* values for our hH_v1-Acd proteins were largely flat across emission wavelengths for our hH_v1-Acd proteins (Figure 5E). For all Acd sites, *Ratio A* was greater than *Ratio A₀* (dashed line in Figure 5E), reflecting FRET between Trp/Tyr and Acd at every site in hH_v1. Most *Ratio A* values were distributed between 0.60 and 0.80, with two clear exceptions: A18Acd (blue) and F247Acd (magenta) (Figure 5E). A18Acd showed the highest *Ratio A*, consistent with the multiple Trp/Tyr residues in the N-terminal domain. Conversely, F247Acd showed the lowest *Ratio A*, consistent with its location in the CC, far from most Trp/Tyr residues (Figure 5B).

We next compared *Ratio A - Ratio A₀* values with FRET efficiencies predicted by the AlphaFold structural model. We modeled rotamer ensembles for Trp, Tyr, and our Acd sites in the hH_v1 structural model using chiLife (36) and calculated donor-acceptor distance distributions. These distributions were then used to calculate FRET efficiencies (details in SI Text). The experimentally determined *Ratio A - Ratio A₀* values showed a moderate correlation (Pearson's $r = 0.48$) with the calculated efficiencies (Figure 5F).

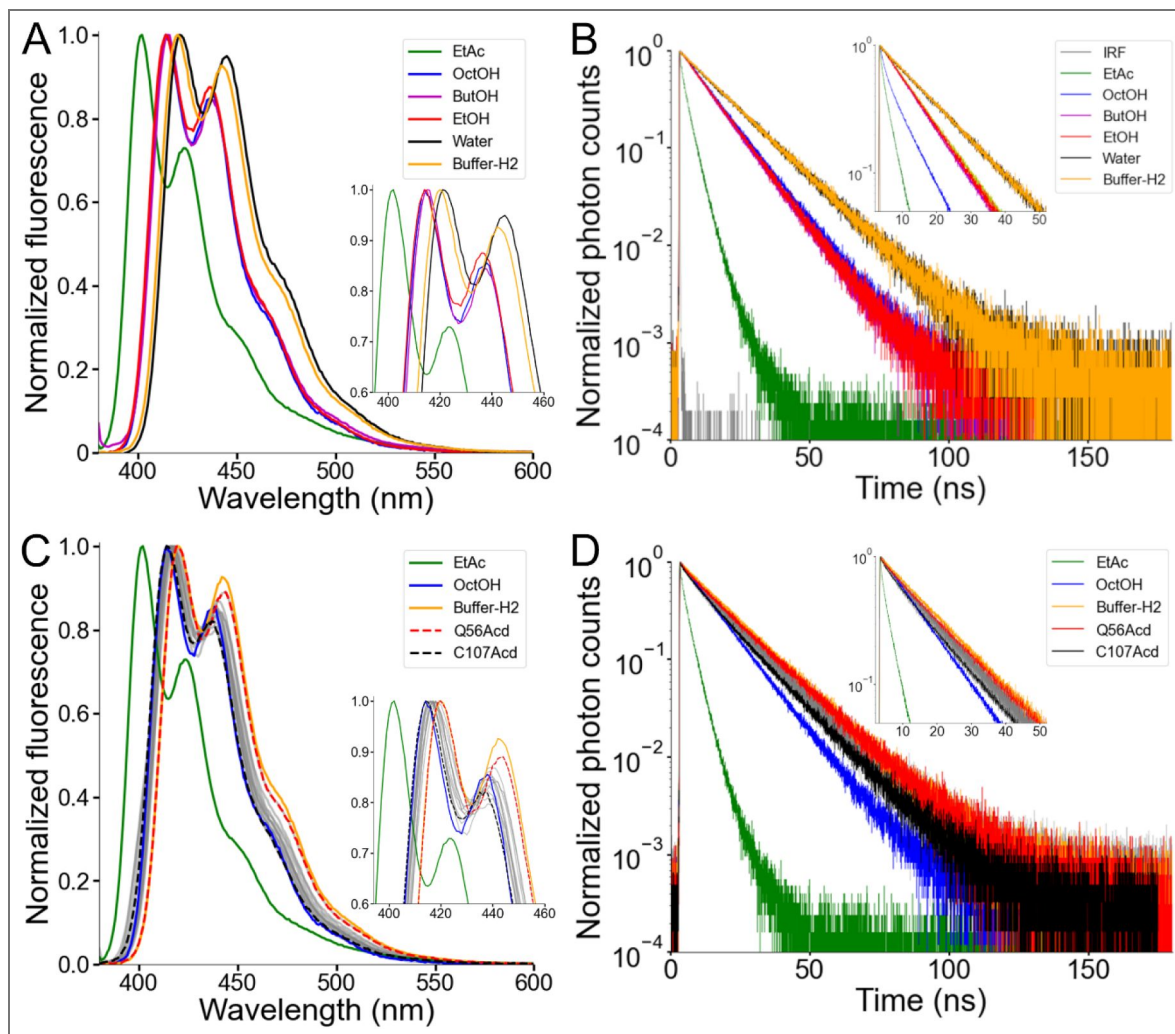


Figure 4. Acd showed a low environmental sensitivity in hH_v1.

(A) Normalized fluorescence emission spectra of the Acd amino acid dissolved in the indicated solvent (Excitation = 370 nm). EtAc: ethyl acetate, OctOH: 1-octanol, ButOH: 1-butanol, EtOH: ethanol. (B) Normalized fluorescence decay of the Acd amino acid dissolved in the indicated solvent. (C) Normalized fluorescence emission spectra of the hH_v1-Acd proteins in Buffer-H2 (grey). The Acd amino acid spectra in selected solvents are also shown for reference. Q56Acd and C107Acd spectra are shown with broken lines. The emission maxima are listed in Table S3. (D) Normalized fluorescence decay of the hH_v1-Acd proteins in Buffer-H2 (grey). The Acd amino acid decays in selected solvents are also shown for reference. Q56Acd and C107Acd decays are shown in red and black, respectively.

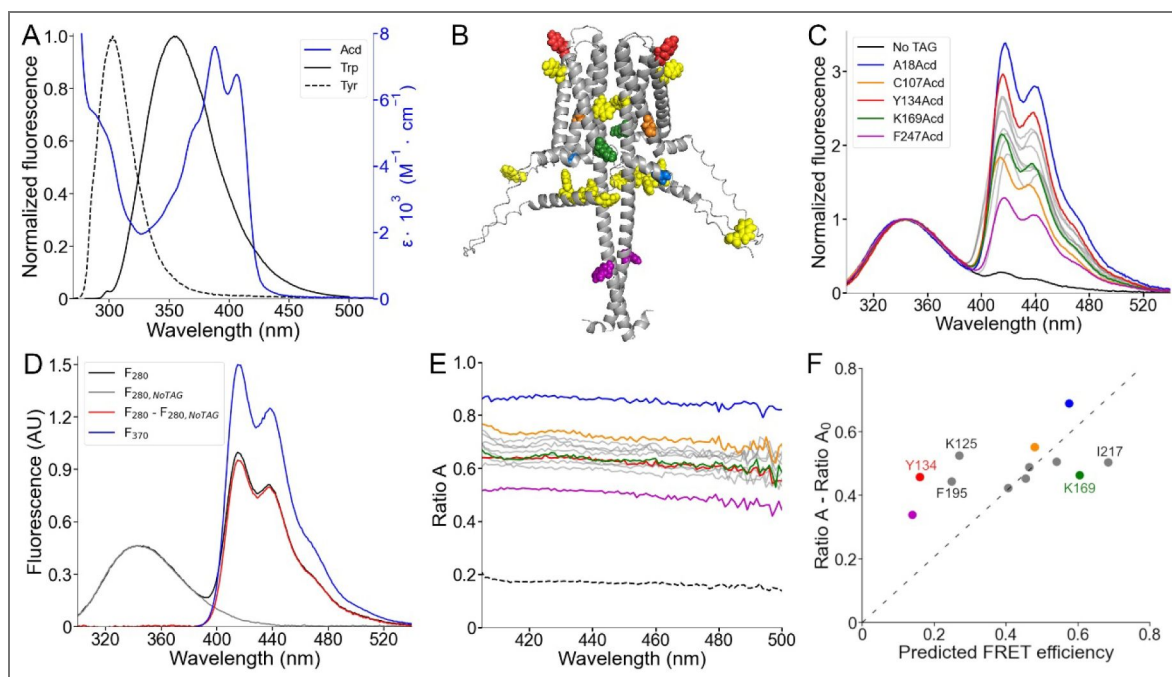


Figure 5. Spectral FRET analysis using Trp and Tyr donors and Acd as acceptor.

(A) Fluorescence emission spectra of the intrinsic fluorescent amino acids Trp (black) and Tyr (black, broken lines), along with the Acd absorption spectrum in Buffer-H2 (blue). (B) AlphaFold dimer hHv1 structural model with the amino acids colored according to (C) and the native tryptophan and tyrosine residues shown as yellow spheres. (C) Normalized fluorescence spectra of the purified hHv1-Acd proteins (Excitation = 280 nm). The spectra were normalized by the Trp/Tyr fluorescence. The colors correspond to the indicated hHv1-Acd protein, and the remaining spectra are shown as grey traces in the figure. The No TAG sample (black) contains purified hHv1 protein without any amber stop codon expressed in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair. (D) Example of the spectral FRET analysis procedure. The fluorescence spectrum of hHv1-Y134Acd obtained when exciting at 280 nm (black, F_{280}) minus the normalized spectrum at the same excitation wavelength of the No TAG sample (grey, $F_{280, NoTAG}$) produced the Acd emission spectrum shown in red ($F_{280} - F_{280, NoTAG}$). Ratio A values were calculated by dividing this spectrum by the hHv1-Y134Acd fluorescence spectrum obtained by direct excitation at 370 nm (F_{370}). (E) Ratio A traces of the hHv1-Acd proteins colored according to (C). The ratio between the spectra of free Acd amino acid excited at 280 and 370 nm (Ratio A₀) is shown as broken lines. (F) Mean Ratio A - Ratio A₀ values (410-480 nm; N=4) as a function of the predicted FRET efficiencies from the AlphaFold hHv1 structural model, colored according to (C). The broken line is the best fit to the equation $(Ratio A - Ratio A_0) = m(Predicted FRET efficiency)$.

There were two spatially clustered deviations between experiment and model. Positions at the extracellular ends of the VSD helices (Y134 in S2, K125 in S1, and F195 in S4) showed higher experimental FRET than predicted, indicating shorter effective donor-acceptor distances relative to the model. Conversely, positions on the intracellular side of the VSD (K169 in S3 and I217 in S4) showed lower experimental FRET than predicted, implying longer effective donor-acceptor distances. Together, the spatial clustering of residues with similar deviations indicates specific deficits in the AlphaFold model or the failure to quantify the protein dynamics, which can guide future improvements.

Zinc binding changed the conformation of hH_v1

We next asked whether FRET from Trp/Tyr to Acd could report conformational changes upon addition of Zn²⁺, the classical inhibitor of H_v1 proton currents in cells (7). The two experimental structures of H_v1 were resolved with this cation bound to the extracellular side of the VSD (11, 12). We observed site-specific spectral changes in the hH_v1-Acd proteins in the presence of Zn²⁺ (Figure 6A and S10). The largest changes occurred for hH_v1-K169Acd, where 1 mM Zn²⁺ caused a decrease in the Trp/Tyr emission and an increase in the Acd emission (Figure 6A). This decrease in donor fluorescence, paired with increased acceptor fluorescence, is the hallmark of increased FRET and was corroborated by spectral analysis (Figure 6B). Additionally, a slight blue shift of the Trp/Tyr fluorescence was observed for all hH_v1-Acd proteins in the presence of Zn²⁺ (Figure S11). These changes were completely reversed by the addition of EDTA, confirming that the changes arise from a reversible conformational change upon Zn²⁺ binding (Figure 6A and 6B). Large changes in effective FRET efficiency were observed with Acd incorporated at Q56 in the N-terminal domain and C107, F159, and K169 in the VSD (Figure 6C). Notably, these positions are intracellular, even though H_v1 currents are inhibited by extracellular Zn²⁺ (7, 37). These data demonstrate that Zn²⁺ binds to hH_v1 and alters its conformation in detergent micelles, consistent with a nuclear magnetic resonance study of the VSD of hH_v1 (12).

Discussion

Incorporating Acd into hH_v1 by GCE opens new possibilities for studying the structure and function of this channel. We systematically incorporated Acd at multiple positions in hH_v1 and confirmed protein stability and function after purification. 12 of 14 hH_v1-Acd proteins produced acceptable yields upon purification and were functional, despite being a human membrane protein expressed in *E. coli*. Two factors likely underlie the robustness of our system: (i) an optimized purification protocol for hH_v1 (20, 21) that offsets the expected reduction in yield associated with GCE (22, 25), and (ii) the high efficiency and specificity of the evolved aminoacyl-tRNA synthetase for incorporating Acd (17, 27).

Fluorescence spectroscopy has been used to detect conformational changes in H_v1. Sites at the extracellular portion of S4 (38) and S1 (39) in the VSD of *Ciona intestinalis* H_v1 have been labeled with the fluorophores Alexa488 and tetramethylrhodamine (TAMRA) through cysteine-selective reagents to capture conformational changes during activation using voltage-clamp fluorometry. Similarly, the fluorescent noncanonical amino acid 3-(6-acetylnaphthalen-2-ylamino)-2-aminopropanoic acid (Anap) has been incorporated at several sites in the S4 in hH_v1 (40) to study conformational changes in response to voltage and pH. In these cases, the fluorophore's environmental sensitivity has been used as a readout of conformational changes. By contrast, FRET provides richer structural information due to its distance dependence. Single-molecule FRET has been measured in purified hH_v1 labeled with Cy3 and Cy5 to track the VSD movements in response to voltage, pH, cholesterol, and zinc (14, 41). The small size and minimal linker of Acd, along with its low environmental sensitivity and high photostability, are advantageous for measuring FRET in hH_v1 compared with previously used fluorophores.

The high number of FRET donors, 4 tryptophans and 4 tyrosines per subunit, limits the utility of the Trp/Tyr-Acd FRET pair as a strategy to measure conformational rearrangements in hH_v1. Our library of a dozen single Acd-incorporating constructs, with sites distributed across the entire

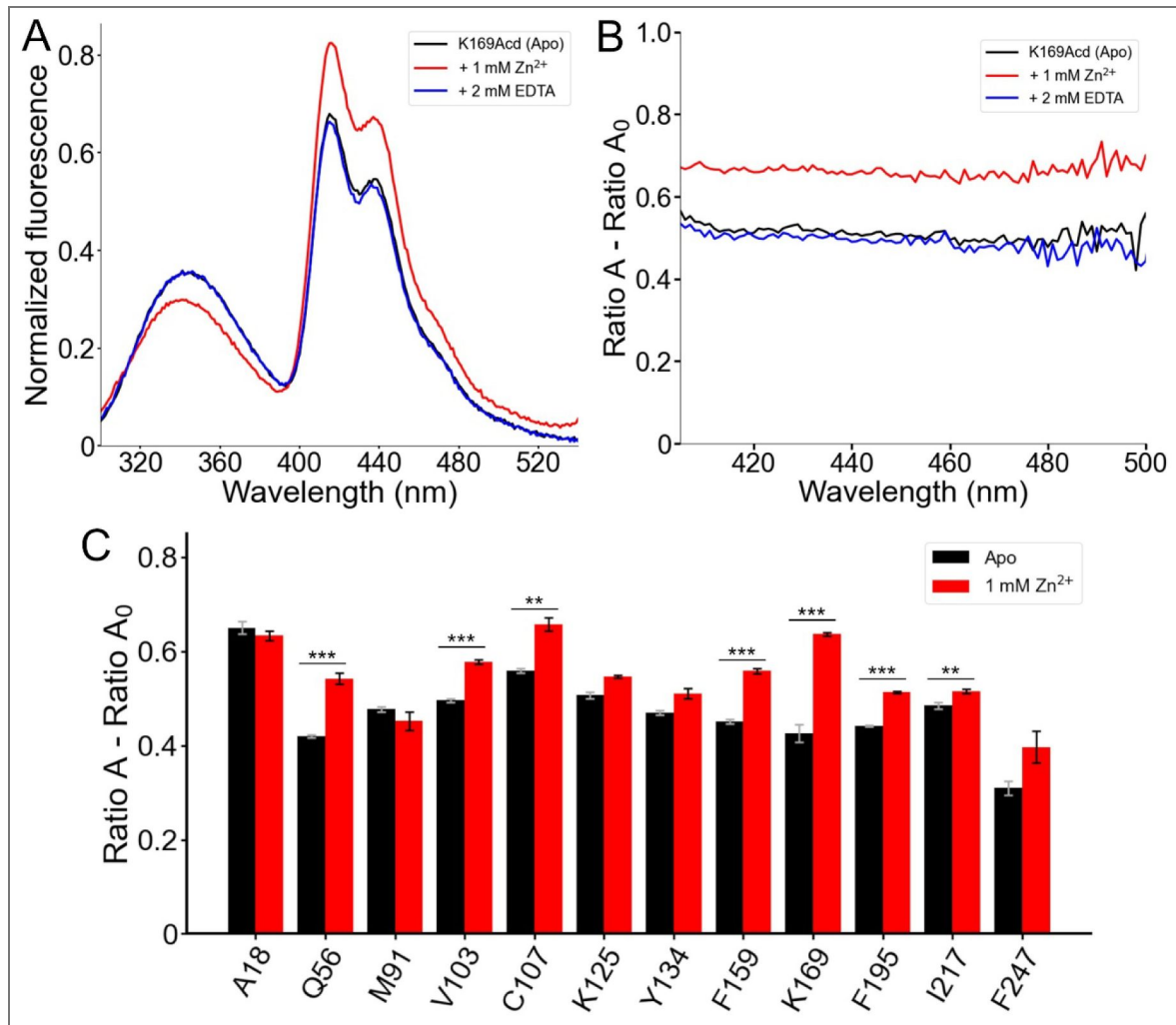


Figure 6. FRET between Trp/Tyr and Acd reports a conformational rearrangement in response to Zn²⁺.

(A) Normalized fluorescence emission spectra of hH_v1-K169Acd (Excitation = 280 nm) in Buffer-H2 in the absence of Zn²⁺ (black, Apo), in the presence of 1 mM Zn²⁺ (red), or in the presence of 1 mM Zn²⁺ and 2 mM EDTA (blue). Spectra were normalized by the maximum intensity of the Acd emission spectrum of the same sample excited at 370 nm. (B) Spectral FRET analysis of hH_v1-K169Acd in Buffer-H2 in the absence of Zn²⁺ (black, Apo), in the presence of 1 mM Zn²⁺ (red), or in the presence of 1 mM Zn²⁺ and 2 mM EDTA (blue). (C) Spectral FRET analysis of the hH_v1-Acd proteins in Buffer-H2 in the absence (black, Apo) or presence of 1 mM Zn²⁺ (red). N=4, ***p<0.005, **p<0.01.

primary sequence of the protein, opens the door to using Acd as a FRET donor. We have previously used time-resolved transition metal ion FRET (tmFRET), which utilizes a transition metal ion as a FRET acceptor (27, 33), to measure conformational energetics in the bacterial cyclic nucleotide-gated ion channel SthK (42). However, in that work, we incorporated Acd at a single position in the intracellular C-terminal domain of SthK. With Acd incorporated at multiple sites in full-length hH_v1, we can interrogate its gating mechanisms in unprecedented detail.

Materials and Methods

Protein expression and purification

hH_v1 was expressed and purified according to the published protocol with minor modifications (20, 21). Briefly, fresh competent BL21-Gold(DE3) cells (Agilent) were co-transformed with the His-EK-hH_v1-C107A-C249A.pET15-b (hH_v1-Cysless, referred to as No TAG) and MjA9Acd-RS.pDule2 (22, 23) plasmids (sequences in SI Text) using the heat shock method. For expressing hH_v1-Acd proteins, the codon corresponding to the selected position was changed to the amber stop codon (TAG) by site-directed mutagenesis. All plasmids were sequenced before use. Pre-cultures were grown overnight in LB medium supplemented with 0.2% glucose, 0.4 mg/mL ampicillin, and 0.1 mg/mL spectinomycin at 37 °C and 250 rpm. The next day, cultures were diluted 1:100 in complete autoinduction medium (43) supplemented with 0.4 mg/mL ampicillin and 0.1 mg/mL spectinomycin, and then grown at 37 °C and 250 rpm until an OD₆₀₀ of 1.5 was reached. The protein expression level was proportional to the Acd concentration in the culture medium (Figure S11B). We chose 0.6 mM Acd in the culture medium as optimal for hH_v1 expression, as higher concentrations only slightly increased the final biomass (Figure S11B). Cultures supplemented with 0.6 mM Acd were grown overnight at 20 °C and 250 rpm. Cells were then harvested, resuspended in Buffer-H1 (50 mM Tris, 150 mM NaCl, 1 mM benzamidine, 0.17 mg/mL PMSF, pH 8.0) supplemented with 0.5 mg/mL lysozyme, and incubated for 30 min at 4 °C before being flash-frozen. After thawing, the extracts were supplemented with 5 mM MgCl₂, fresh protease inhibitors, and 12.5 µg/mL DNase I, incubated for 1 h at 4 °C, sonicated, and centrifuged at 100,000g at 4 °C for 1 h. The membrane pellet was resuspended in Buffer-H1 and stored at -80 °C until further use. After thawing, the membranes were solubilized with 1.5% Anzergent 3-12 (Anatrace) (Anz3-12) at room temperature for 1 hour. Insoluble material was removed by centrifuging at 100,000g and 4 °C for 1 h, and the solubilized membrane extract was incubated with His-Pur Ni-NTA (Thermo Scientific) resin equilibrated with Buffer-H2 (50 mM Tris, 150 mM NaCl, 12 mM Anz3-12, pH 8.0) for 1 hour at room temperature. Resin was collected in a gravity column and washed with 20 column volumes (CV) of Buffer-H2, followed by 16 CV of Buffer-H2 with 90 mM imidazole. The hH_v1 protein was eluted with 20 CV of Buffer-H2 with 0.5 M imidazole, was concentrated with 50kDa cut-off centrifugal filters, and imidazole was removed by FSEC (Ex/Em = 385/450 nm) in an ENrich SEC 650 (Bio-Rad) column with Buffer-H2. The hH_v1-containing fractions were concentrated to 0.5-1.5 mg/mL, depending on the amount of purified protein, and then aliquoted, flash-frozen, and stored at -80 °C until use.

Reconstitution and fluorescence proton flux assays

The hH_v1-Acd samples were reconstituted and assayed using a liposome proton flux assay, as described previously, with minor modifications (20). Briefly, soy polar lipid extract (Avati) in chloroform was dried overnight at room temperature in a rotary evaporator under vacuum and then hydrated in Buffer-K (20 mM HEPES, 150 mM KCl, and 1 mM EDTA, pH 7.0) to a concentration of 10 mg/mL. The vesicles were sonicated in a water bath until the solution became translucent, aliquoted, flash-frozen, and stored at -80 °C until further use. Reconstitutions were performed at a 1:100 (protein:lipid, by mass) ratio by mixing 12.5 µg of purified hH_v1-Acd with 1.25 mg of liposomes in a final volume of 330 µL Buffer-K containing 8 mM Anz3-12. The mixture was incubated at room temperature for 1 h, then diluted with 5 mL of Buffer-K and incubated for 30 min. The mixture was further incubated with three cumulative pulses of 100 mg Bio-Beads SM2 (Bio-Rad), each for 1 h at room temperature. After the final Bio-Beads pulse, the mixture was incubated overnight at 4 °C. The next day, two successive 100 mg additions of Bio-Beads were

performed, with a 1-hour incubation at room temperature following each addition. Bio-Beads were removed in a gravity column, the mixture was diluted with 10 mL of Buffer-K, and then centrifuged at 150,000g for 2 hours at 4 °C. The proteoliposome pellets were resuspended in 250 μ L of Buffer-K, aliquoted, flash-frozen, and stored at -80 °C until further use. On the day of the liposome proton flux assays, samples were thawed at 37 °C and then placed on ice. The sample (40 μ L) was diluted in Buffer-Na (20 mM HEPES, 150 mM NaCl, 1 mM EDTA, pH 7.0). 9-Amino-6-chloro-2-methoxyacridine (ACMA; Sigma) was added to a final concentration of 2 μ M from a 2 mM stock solution in DMSO, and the mixture was incubated at room temperature for 5 minutes before the fluorescence measurement began. The baseline was recorded for 3 min (F_{max}). Subsequently, 10 nM of valinomycin (Cell Signaling) was added from a 10 μ M stock solution in DMSO, followed by 1 μ M carbonyl cyanide m-chlorophenyl hydrazone (CCCP; Sigma) from a 1 mM stock solution in DMSO to record F_{min} . The final volume was 2 mL. The fluorescence signal ($F(t)$) was normalized according to the equation $(F(t)-F_{min})/(F_{max}-F_{min})$. Fluorescence measurements were performed at 5-second intervals with a 2-second integration time using a Fluorolog-3 spectrometer (Horiba) configured to an excitation wavelength of 410 nm (5 nm bandpass) and an emission wavelength of 490 nm (1 nm bandpass) at room temperature.

Absorption and fluorescence spectra recordings

The absorption and fluorescence spectra were recorded in the indicated solvents or fresh Buffer-H2 at room temperature. Absorption spectra were recorded in a DU 800 Spectrometer (Beckman Coulter) with a wavelength interval of 0.5 nm and a scan speed of 600 nm/min. Fluorescence emission spectra were measured using a Fluorolog-3 spectrometer (Horiba) with an integration time of 0.1 s, 1 nm increments, and excitation and emission slits of 5 nm. For solubilizing Acd in different solvents, a saturated solution of the amino acid was prepared by dissolving 0.5 mg of Acd powder in the respective solvent. Acd spectra were corrected by subtracting a blank of the solvent. hH_v1-Acd proteins were measured at concentrations ranging from 180 to 240 nM in Buffer-H2. Each spectrum was corrected by subtracting a blank with the solvent or Buffer-H2 before adding the protein sample to the cuvette.

Spectral FRET analysis

We used our previously established spectral FRET analysis to remove contamination caused by direct excitation of Acd by 280 nm light (35). This method had the added benefit of eliminating errors arising from the recording system's transfer function, variations in the acceptor's quantum yield, or variations in the total concentration of fluorescent molecules. A Trp/Tyr spectrum was collected from control protein without any TAG codons (no Acd incorporation). This was used to subtract the Trp/Tyr emission spectra collected at 280 nm for each Acd-incorporating protein. This yielded the extracted Acd spectrum, F_{280} , that had two components: the component due to direct excitation of Acd, F_{280}^{direct} , and the component due to FRET, F_{280}^{FRET} . F_{280} was normalized by the total Acd emission with 370 nm excitation, F_{370} . The resulting ratio, termed *Ratio A*, can be expressed as: $Ratio A = \frac{F_{280}}{F_{370}} = \frac{F_{280}^{direct}}{F_{370}} + \frac{F_{280}^{FRET}}{F_{370}}$. The direct excitation component, $\frac{F_{280}^{direct}}{F_{370}}$, termed *Ratio A₀*, was measured using free Acd amino acid in Buffer-H2. We then quantified the relative FRET efficiency as $Ratio A - Ratio A_0 = \frac{F_{280}^{FRET}}{F_{370}}$. This quantity is directly proportional to FRET efficiency. An average *Ratio A* value was calculated between 410 and 480 nm.

Time-resolved fluorescence measurements

TCSPC measurements were performed in a FluoTime 300 spectrometer equipped with a PMA Hybrid 40 detector and LDH-P-C-375 laser diode head (PicoQuant) at room temperature. Emitted photons were detected at 446 nm with the emission polarizer at the magic angle. The TCSPC measurements were performed in the indicated solvent or fresh Buffer-H2. The hH_v1-Acd samples were measured at a concentration of 1 μ M and the Acd amino acid at 500 nM. The count rate was always less than 1% of the excitation repetition rate to avoid pile-up distortions of the fluorescence decay. The IRF was measured using a dilute Ludox solution in water.

Supplementary Text

Calculation of the FRET efficiency from the structural model

We calculated the FRET efficiency using the AlphaFold model of the full-length hHv1 (Dataset S1). For each Acd position, the measured steady-state FRET would correspond to the mean FRET efficiency (E_{FRET}), which will be proportional to the *Ratio A – Ratio A₀* value obtained from the spectral FRET analysis.

For n number of donors and two acceptors in the hHv1 dimer:

$$E_{FRET} = \sum_{i=1}^n P_{Ex,i} E_i \quad (1)$$

where $P_{Ex,i}$ is the probability of excitation and E_i is the FRET efficiency of the i th donor. $P_{Ex,i}$ is the number of photons absorbed by the i th donor divided by the total number of photons absorbed by the total number n of donors, which is quantified by the donor's extinction coefficient. Supposing the same extinction coefficient for each of the Trp ($\epsilon_W = 5,501 \text{ M}^{-1} \text{ cm}^{-1}$) and Tyr ($\epsilon_Y = 1,209 \text{ M}^{-1} \text{ cm}^{-1}$) (44), equation 1 (1) can be written as:

$$E_{FRET} = \sum_{i=1}^n \left(\frac{\epsilon_i}{\sum_{i=1}^n \epsilon_i} \right) E_i = \frac{1}{n_W \epsilon_W + n_Y \epsilon_Y} (\epsilon_W \sum_{i=1}^{n_W} E_{W,i} + \epsilon_Y \sum_{i=1}^{n_Y} E_{Y,i}) \quad (2)$$

where the sum of the total number of Trp (n_W) and Tyr (n_Y) is equal to n . The efficiency of FRET for the i th donor can be expressed as the sum of rates of energy transfer to each of the $j = 2$ Acd acceptors, k_{ij} , divided by the total rates of photon emission. Therefore, the right terms of equation 2 (2) can be expressed as:

$$\sum_{i=1}^{n_W} E_{W,i} = \sum_{i=1}^{n_W} \left(\frac{\sum_{j=1}^2 k_{ij}}{\sum_{j=1}^2 k_{ij} + (1/\tau_W)} \right) \quad \text{and} \quad \sum_{i=1}^{n_Y} E_{Y,i} = \sum_{i=1}^{n_Y} \left(\frac{\sum_{j=1}^2 k_{ij}}{\sum_{j=1}^2 k_{ij} + (1/\tau_Y)} \right) \quad (3)$$

where the Trp lifetime ($\tau_W = 3.1 \text{ ns}$) and tyrosine lifetime ($\tau_Y = 3.6 \text{ ns}$) in the absence of donors are considered constant (1). Finally, each energy transfer rate k_{ij} is a function of the distance r_{ij} between the i th donor and the j th acceptor:

$$k_{ij}(r) = \frac{1}{\tau_D} \left(\frac{R_0}{r_{ij}} \right)^6 \quad (4)$$

where the donor lifetime τ_D and R_0 would be τ_W and $23.5 \text{ \AA}$ or τ_Y and $20.9 \text{ \AA}$ for Trp or Tyr residues, respectively. Calculating E_{FRET} using the above equations considers a fixed distance between donors and acceptors. A better approach is to consider the distance distributions $P_{ij}(r)$ produced by the different rotameric states of the FRET pair, which can be modeled easily using chiLife (45). Then, equation 4 (4) will be replaced by equation 5 (5):

$$k_{ij}(r) = \int_0^\infty P_{ij}(r) \left[\frac{1}{\tau_D} \left(\frac{R_0}{r} \right)^6 \right] dr \quad (5).$$

DNA plasmid sequences in FASTA format

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Supplementary Figures

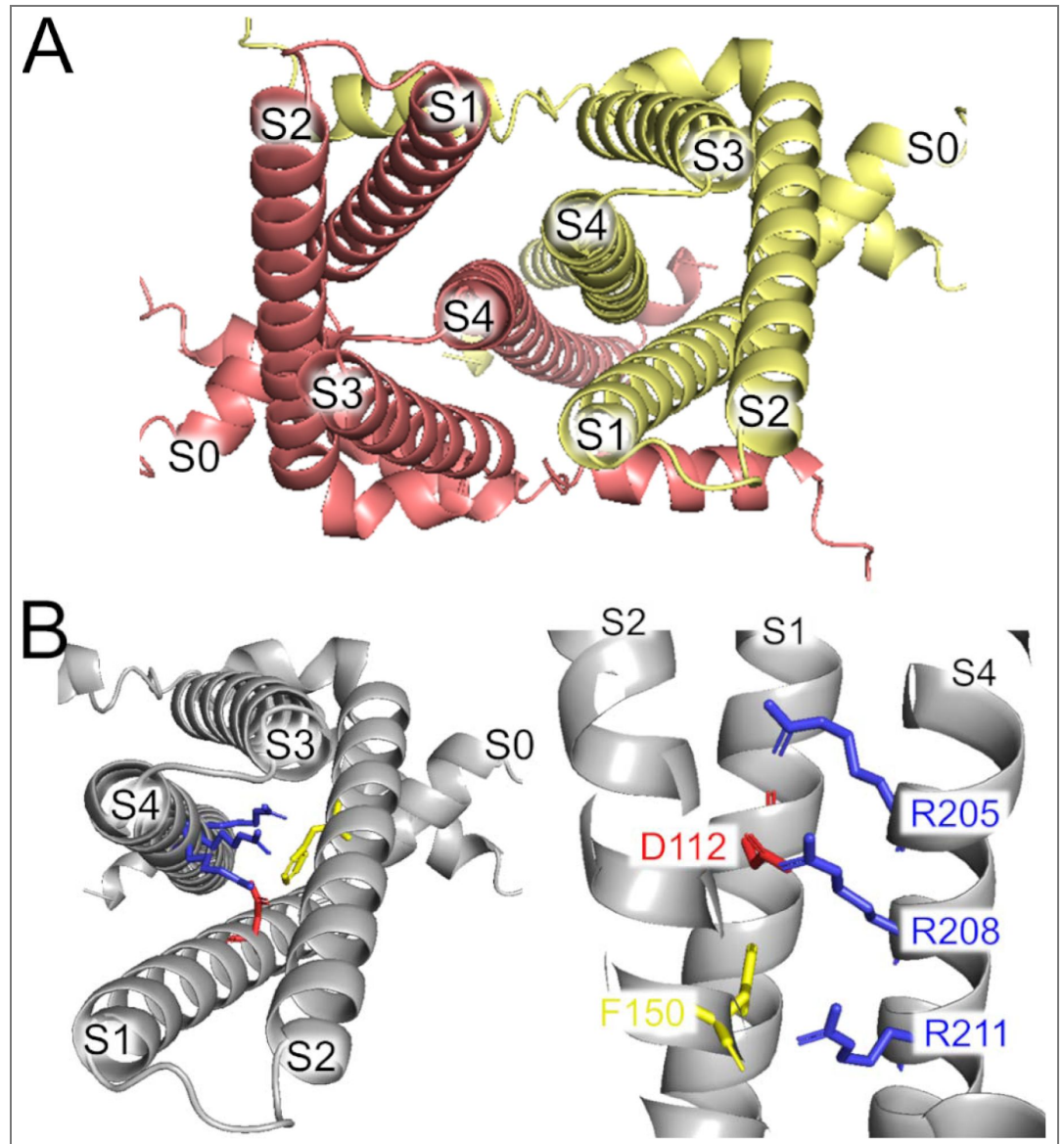


Figure S1. AlphaFold dimer hHv1 structural model. (A) Extracellular view of the model, showing each subunit in a different color. (B) One subunit of the dimer showing the position of the arginine gating charges in S4 (blue), the selectivity filter (red), and the charge transfer center (yellow).

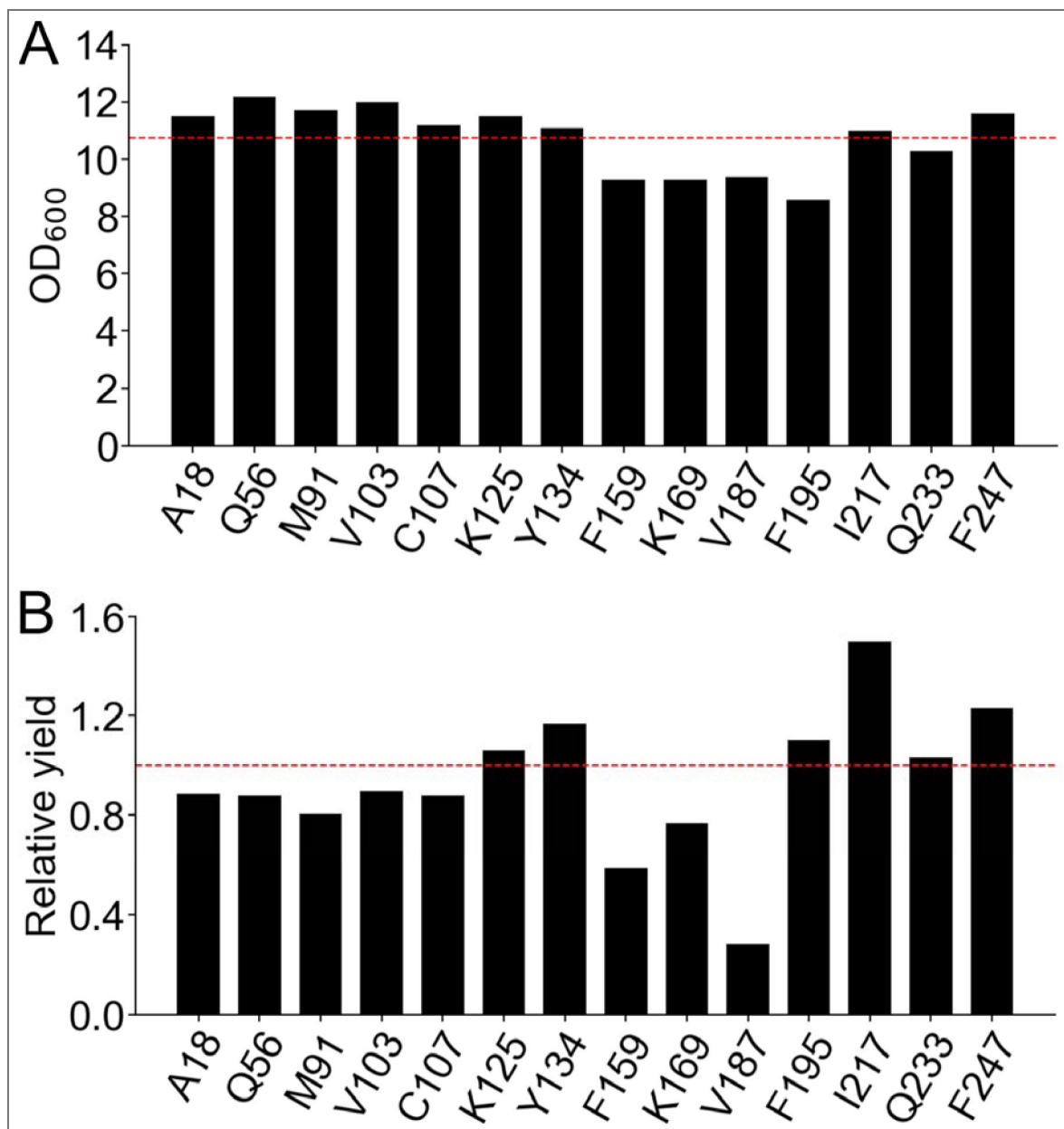


Figure S2. The protein yield of hH_v1-Acd depended on the position of Acd incorporation.

(A) Optical density at 600 nm (OD₆₀₀) of the final cultures expressing the hH_v1 with Acd replacing the indicated amino acid. (B) Relative protein yield calculated by multiplying the OD₆₀₀ and Western blot band intensity of cellular extracts from the final cultures expressing the hH_v1 with Acd replacing the indicated amino acid. Values were normalized by the No TAG condition, which are cells expressing the hH_v1 without any amber stop codon in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair (red broken line).

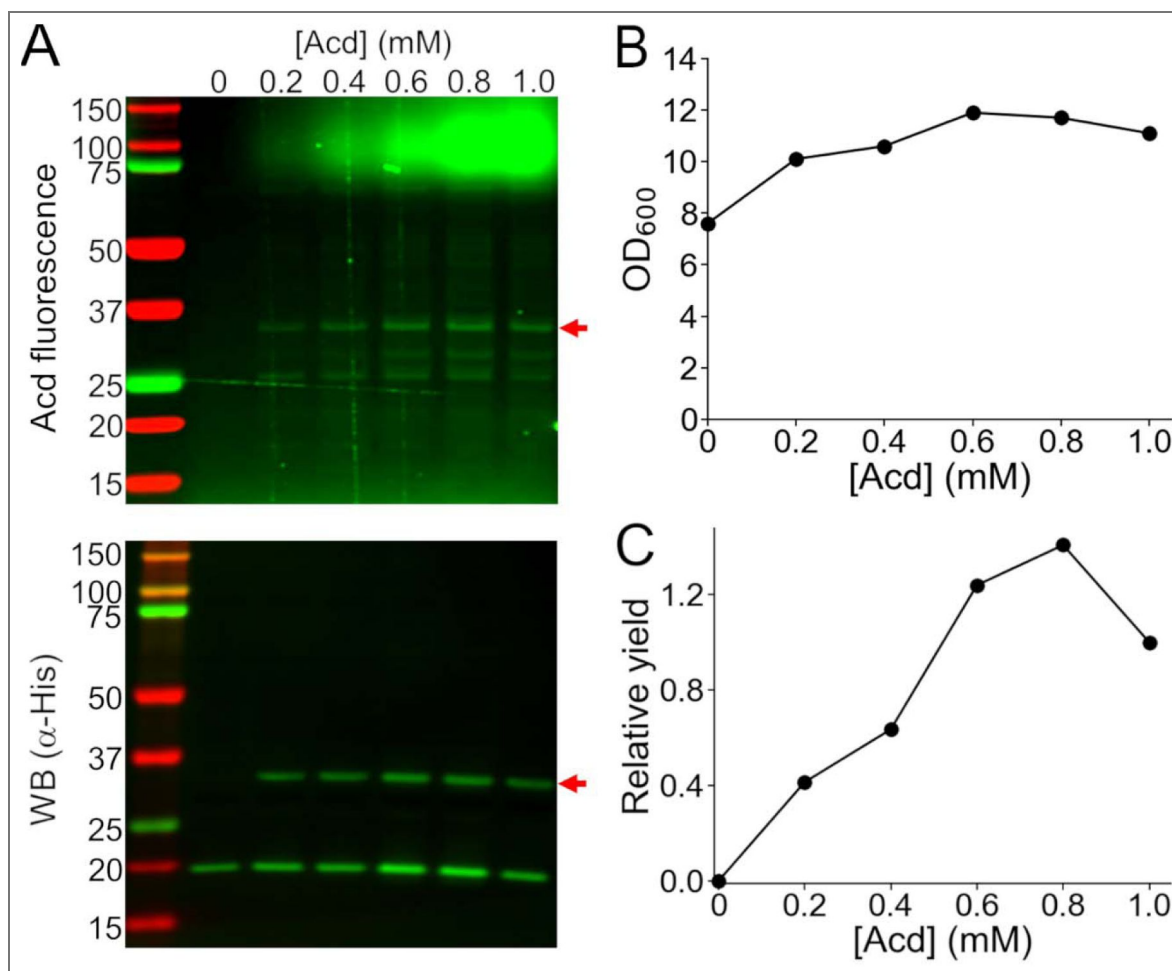


Figure S3. The truncation of hH_v1-K125Acd was not decreased at higher concentrations of Acd in the culture medium.

(A) Cellular extracts of cells co-transformed with the MjA9-AcdRS/tRNA pair plasmid and hH_v1-K125TAG and grown with the indicated concentration of Acd in the culture medium were separated by SDS-PAGE to visualize the hH_v1 band (red arrow) by Acd fluorescence (top) and Western blot against the N-terminus His-tag (bottom). (B) Optical density at 600 nm (OD₆₀₀) of the cultures at the end of the expression. (C) Relative yield obtained by multiplying the Western blot band intensity by the OD₆₀₀, normalized by the value obtained at 1 mM Acd.

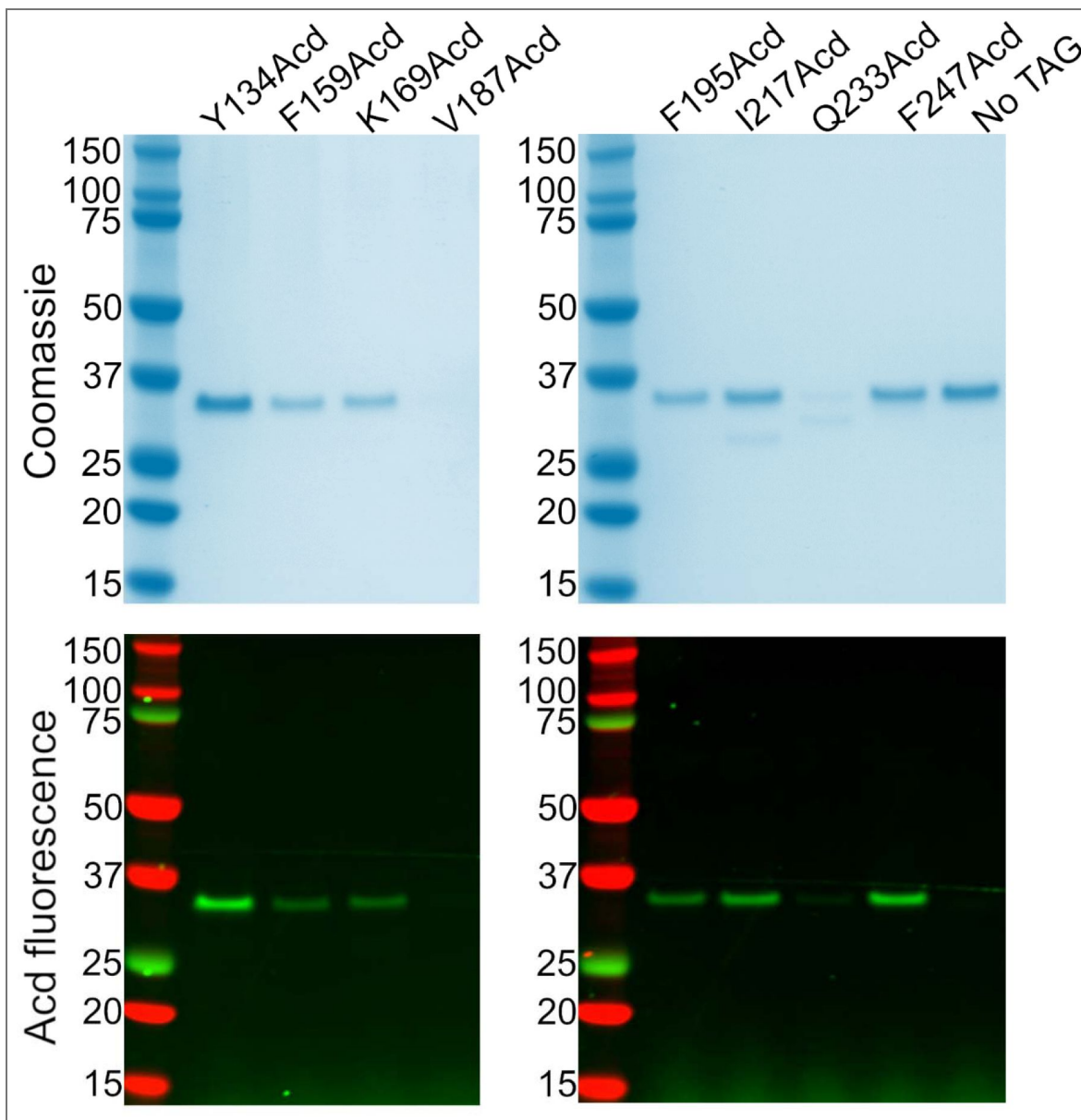


Figure S4. SDS-PAGE of the hHv1 protein samples after purification.

Coomassie-stained (top) and Acd fluorescence (bottom) gels of the samples from the elution of the nickel resin column, separated by SDS-PAGE. Note that V187Acd and Q233Acd did not contain appreciable amounts of protein. The No TAG condition consisted of samples purified from cells expressing the hHv1 without any amber stop codon in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair.

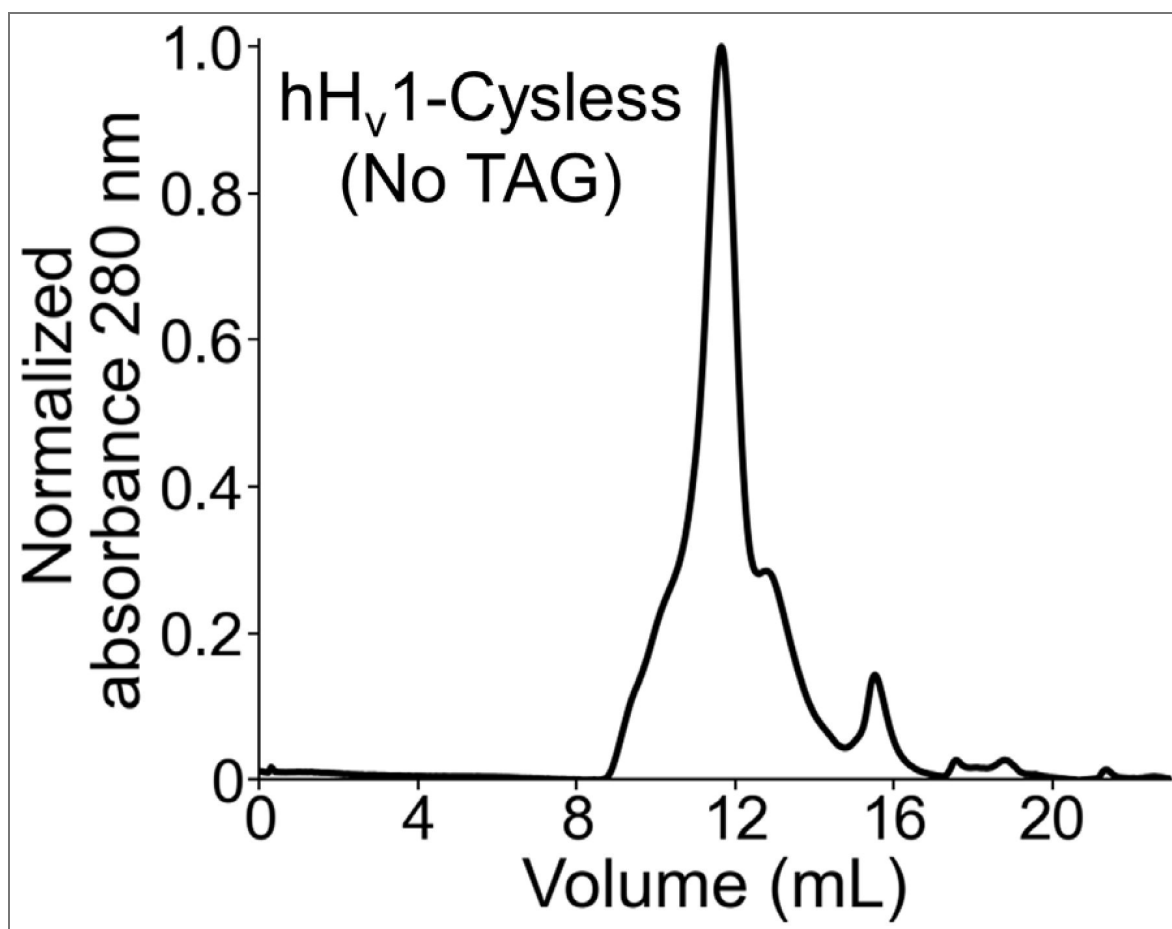


Figure S5. Size exclusion chromatogram of the No TAG sample.

The No TAG sample was purified from cells expressing the hH_v1 without any amber stop codon in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair. The sample eluted from the immobilized affinity chromatography column was concentrated and injected into the size exclusion chromatography column.

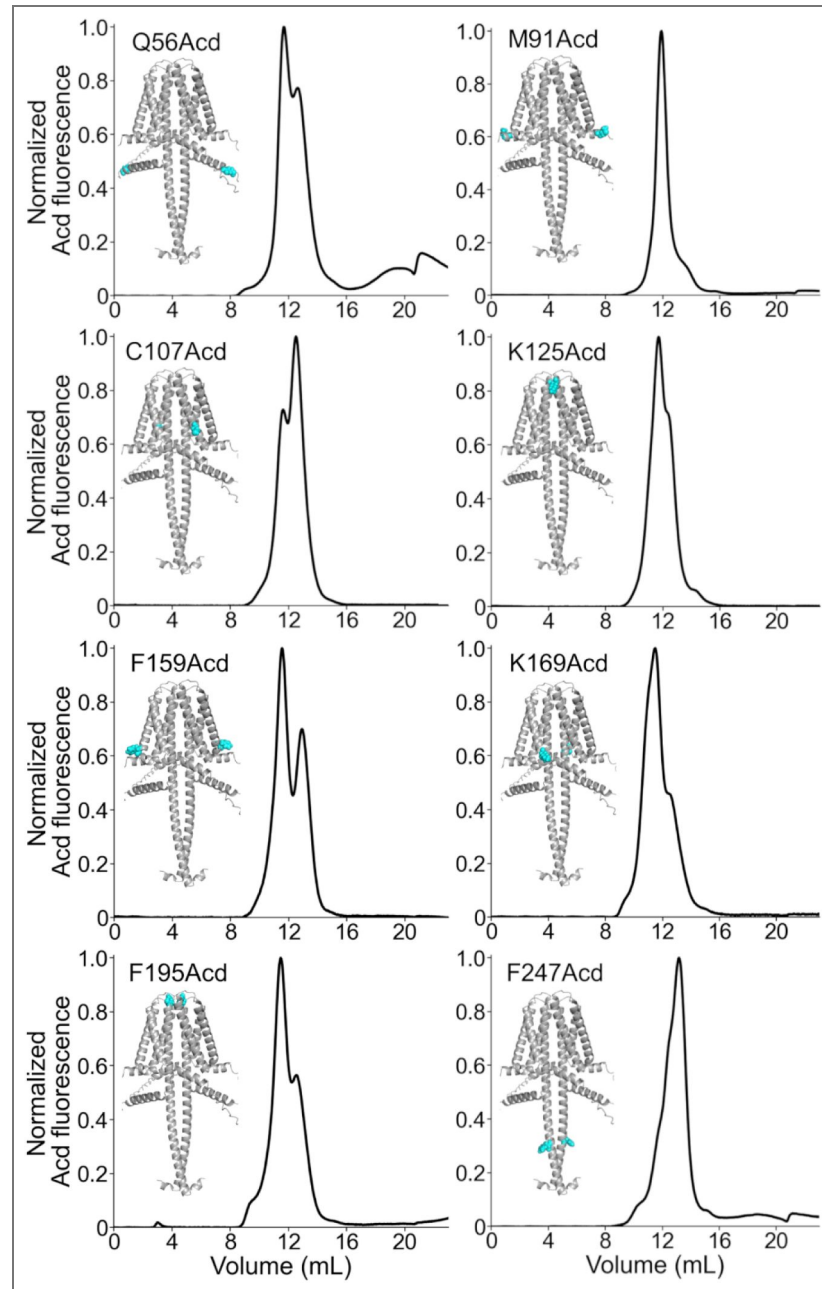


Figure S6. Fluorescence-detection size exclusion chromatography of hHv1-Acd proteins.

Chromatograms of the purified hHv1 proteins with Acd incorporated at the indicated amino acid position. The AlphaFold dimer model with the amino acid replaced by Acd as cyan spheres is included.

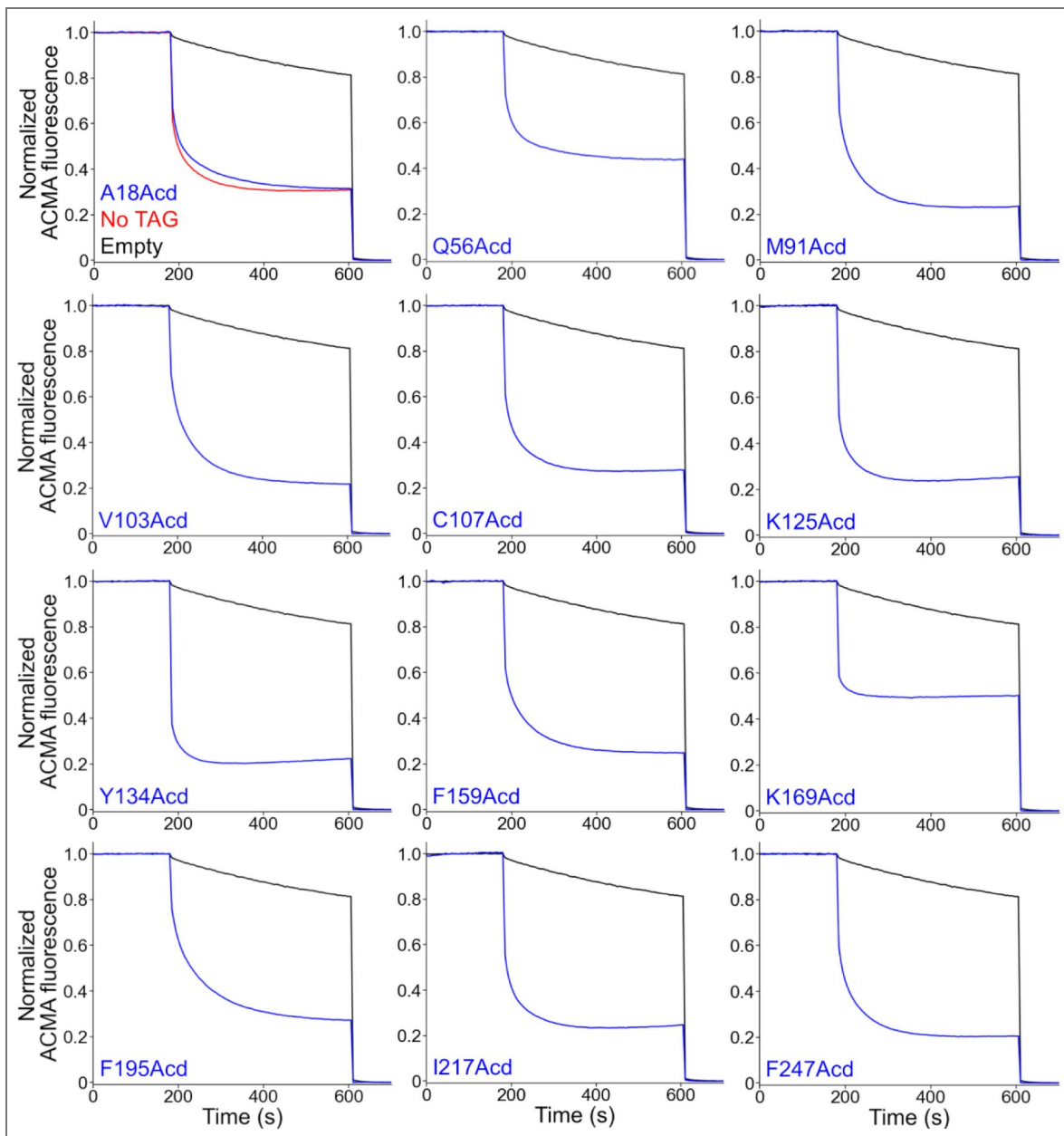


Figure S7. The purified hH_v1-Acd proteins were functional proton channels.

Liposome proton flux assays of asolectin proteoliposomes containing the indicated hH_v1-Acd protein. The proteoliposome samples produced ACMA fluorescence quenching after the addition of valinomycin (180 s) in the presence of a potassium gradient. The protonophore CCCP was added at the end of the experiment (600 s). The No TAG sample (red trace) contains proteoliposomes containing hH_v1 without any amber stop codon expressed in the presence of Acd and the aminoacyl-tRNA synthetase/tRNA pair. Empty asolectin liposomes (black traces) showed slow ACMA fluorescence quenching.

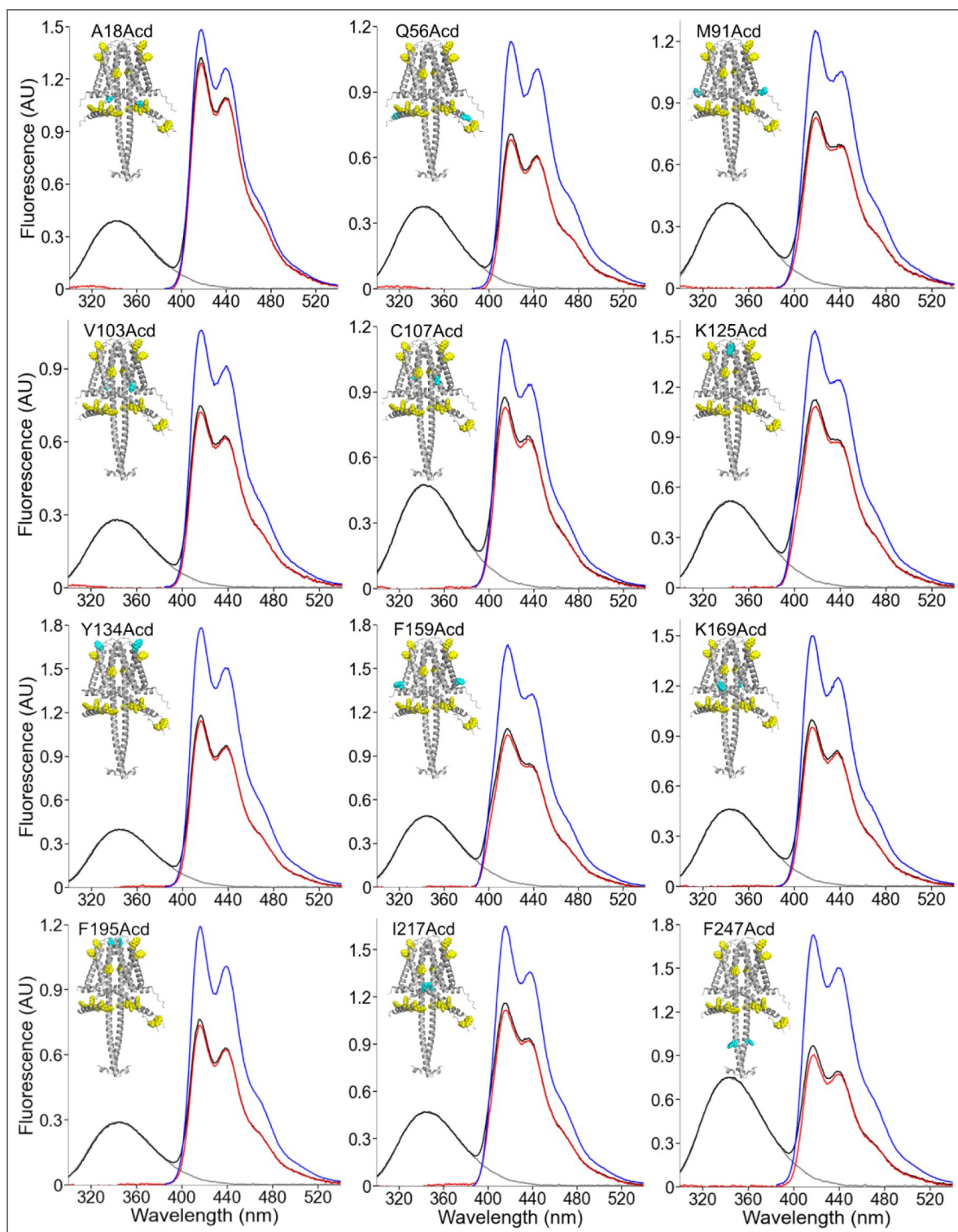


Fig. S8. Fluorescence emission spectra of hHv1-Acd.

Spectra of the indicated hHv1-Acd protein sample obtained when exciting at 280 nm (black) and 370 nm (blue). The isolated spectrum of Acd (red) was obtained by subtracting the black trace from the normalized spectrum of the No TAG sample excited at 280 nm (grey).

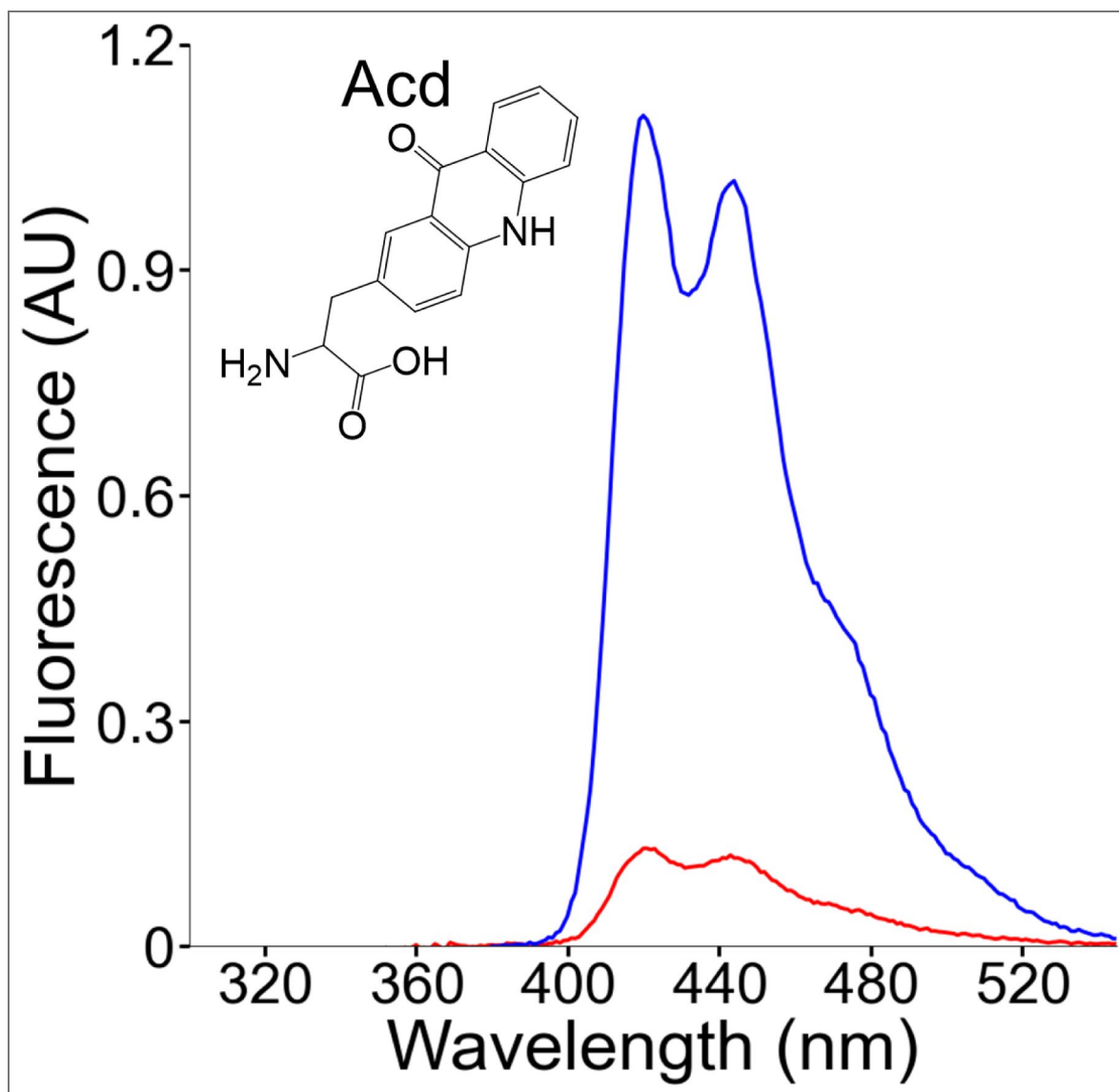


Figure S9. Fluorescence emission spectra of the Acd amino acid in solution.

Spectra of Acd in Buffer-H2 using an excitation wavelength of 370 nm (blue) or 280 nm (red).

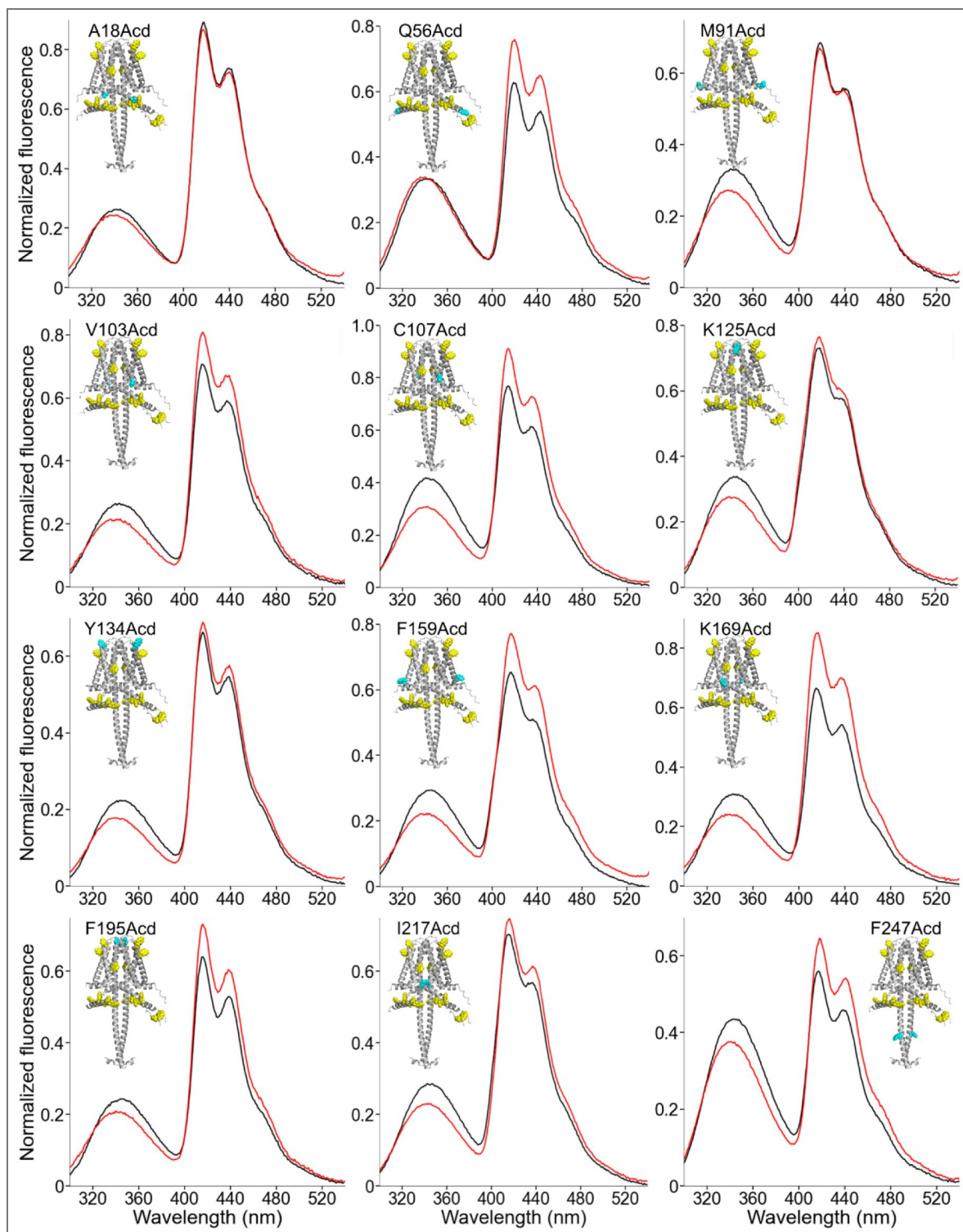


Fig. S10. Changes in the fluorescence emission spectra of hHv1-Acd in the presence of zinc.

Spectra of the indicated hHv1-Acd protein sample obtained when exciting at 280 nm in the Apo state (black) and in the presence of 1 mM Zn^{2+} (red). Spectra were normalized by the maximum intensity of the Acd emission spectrum of the same sample excited at 370 nm.

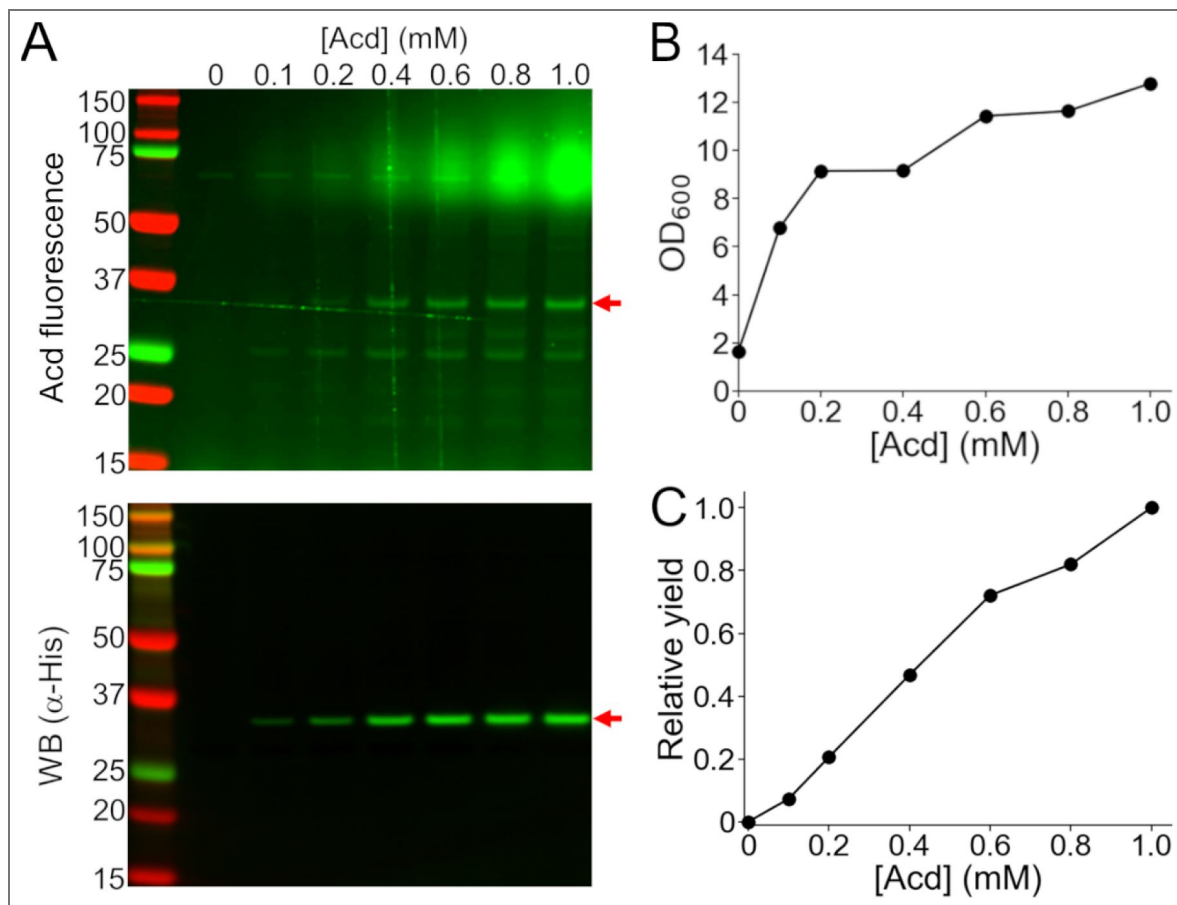


Figure S11. The protein yield of hHv1-A18Acid was proportional to the concentration of Acid in the culture medium.

(A) Cellular extracts of cells co-transformed with the MjA9-AcidRS/tRNA pair plasmid and hHv1-A18TAG and grown with the indicated concentration of Acid in the culture medium were separated by SDS-PAGE to visualize the hHv1 band (red arrow) by Acid fluorescence (top) and Western blot against the N-terminus His-tag (bottom). (B) Optical density at 600 nm (OD₆₀₀) of the cultures at the end of the expression. (C) Relative yield obtained by multiplying the Western blot band intensity by the OD₆₀₀, normalized by the value obtained at 1 mM Acid.

Supplementary Tables

hH _v 1 sample	Elution volume (mL)
No TAG	11.5
A18Acd	11.6
Q56Acd	11.7
M91Acd	11.9
V103Acd	11.5
C107Acd	12.5
K125Acd	11.7
Y134Acd	11.5
F159Acd	11.5
K169Acd	11.5
F195 Acd	11.5
I217Acd	12.4
F247Acd	13.1

Table S1. Size-exclusion chromatography elution volumes of the hH_v1 proteins. Elution volumes were measured as the location of the maximum value of absorbance at 280 nm in the chromatograms.

Solvent	Emission Maximum (nm)
EtAc	402
OctOH	414
ButOH	415
EtOH	414
Water	421
Buffer-H2	420

Table S2. Emission fluorescence spectrum maximum of Acd in different solvents.

The wavelength corresponds to the location of the maximum intensity of the emission fluorescence spectrum of the free amino acid dissolved in the corresponding solvent. EtAc: ethyl acetate, OctOH: 1-octanol, ButOH: 1-butanol, EtOH: ethanol.

hH _v 1 sample	Emission Maximum (nm)
A18Acd	418
Q56Acd	420
M91Acd	419
V103Acd	417
C107Acd	415
K125Acd	418
Y134Acd	416
F159Acd	418
K169Acd	415
F195 Acd	416
I217Acd	415
F247Acd	418

Table S3. The emission fluorescence spectrum maximum of the hH_v1-Acd proteins.

The wavelength corresponds to the location of the maximum intensity of the emission fluorescence spectrum of the samples.

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Additional information

Data availability

All data generated or analyzed during this study are included in the manuscript and supporting files; source data files have been provided for all figures.

Author Contributions:

All the authors conceived the study and designed the experiments. E.M.C. performed the experiments and analyzed the data. All the authors interpreted the data. E.M.C. wrote the first version of the manuscript. All the authors critically revised the article.

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Additional files

Supplementary Dataset S1. [Coordinates of the AlphaFold predicted model of the hH_v1 dimer \(in PDB format\).](#)

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Peer reviews

Reviewer #1 (Public review):

In this study, the noncanonical amino acid acridon-2-ylalanine (Acd) was inserted at various positions within the human Hv1 protein using a genetic code expansion approach. The purified mutants with incorporated fluorophore were shown to be functional using a proton flux assay in proteoliposomes. FRET between native tryptophan and tyrosine residues and Acd were quantified using spectral FRET analysis. Predicted FRET efficiencies calculated from an AlphaFold model of the Hv1 dimer were compared to the corresponding experimental values. Spectral FRET analysis was also used to test whether structural rearrangements caused by Zn^{2+} , a well-known Hv1 inhibitor, could be detected. The experimental data provide a good validation of the approach, but further expansion of the analysis will be necessary to differentiate between intra- and intersubunit structural features.

Interestingly, the observed rearrangements induced by Zn^{2+} were not limited to the protein region proximal to the extracellular binding site but extended to the intracellular side of the channel. This finding agrees with previous studies showing that some extracellular Hv1 inhibitors, such as Zn^{2+} or AGAP/W38F, can cause long-range structural changes propagating to the intracellular vestibule of the channel (De La Rosa et al. J. Gen. Physiol. 2018, and Tang et al. Brit J. Pharm 2020). The authors should consider adding these references.

Since one of the main goals of this work was to validate Acd incorporation and the spectral FRET analysis approach to detect conformational changes in hHv1 in preparation for future studies, the authors should consider removing one subunit from their dimer model, recalculating FRET efficiencies for the monomer, and comparing the predicted values to the experimental FRET data. This comparison could support the idea that the reported FRET measurements can inform not only on intrasubunit structural features but also on subunit organization.

<https://doi.org/10.7554/eLife.110161.1.sa1>

Reviewer #2 (Public review):

This manuscript by Carmona, Zagotta, and Gordon is generally well-written. It presents a crude and incomplete structural analysis of the voltage-gated proton channel based on measured FRET distances. The primary experimental approach is Förster Resonance Energy Transfer (FRET), using a fluorescent probe attached to a noncanonical amino acid. This strategy is advantageous because the noncanonical amino acid likely occupies less space than conventional labels, allowing more effective incorporation into the channel structure.

Fourteen individual positions within the channel were mutated for site-specific labeling, twelve of which yielded functional protein expression. These twelve labeling sites span discrete regions of the channel, including P1, P2, S0, S1, S2, S3, S4, and the dimer-connecting coiled-coil domain. FRET measurements are achieved using acridon-2-ylalanine (Acd) as the acceptor, with four tryptophan or four tyrosine residues per monomer serving as donors. In addition to estimating distances from FRET efficiency, the authors analyze full FRET spectra and investigate fluorescence lifetimes on the nanosecond timescale.

Despite these strengths, the manuscript does not provide a clear explanation of how channel structure changes during gating. While a discrepancy between AlphaFold structural predictions and the experimental measurements is noted, it remains unclear whether this mismatch arises from limitations of the model or from the experimental approach. No further structural analysis is presented to resolve this issue or to clarify the conformational states of the protein.

The manuscript successfully demonstrates that Acd can be incorporated at specific positions without abolishing channel function, and it is noteworthy that the reconstituted proteins function as voltage-activated proton channels in liposomes. The authors also report reversible zinc inhibition of the channel, suggesting that zinc induces structural changes in certain channel regions that can be reversed by EDTA chelation. However, this observation is not explored in sufficient depth to yield meaningful mechanistic insight.

Overall, while the study introduces an interesting labeling strategy and provides valuable methodological observations, the analysis appears incomplete. Additional structural interpretation and mechanistic insight are needed.

Major Points

- (1) Tryptophan and tyrosine exhibit similar quantum yields, but their extinction coefficients differ substantially. Is this difference accounted for in your FRET analysis? Please clarify whether this would result in a stronger weighting of tryptophan compared to tyrosine.
- (2) Is the fluorescence of acridon-2-ylalanine (Acd) pH-dependent? If so, could local pH variations within the channel environment influence the probe's photophysical properties and affect the measurements?
- (3) Several constructs (e.g., K125Tag, Y134Tag, I217Tag, and Q233Tag) display two bands on SDS-PAGE rather than a single band. Could this indicate incomplete translation or premature termination at the introduced tag site? Please clarify.
- (4) In Figure 5F, the comparison between predicted FRET values and experimentally determined ratio values appears largely uninformative. The discussion on page 9 suggests either an inaccurate structural model or insufficient quantification of protein dynamics. If the underlying cause cannot be distinguished, how do the authors propose to improve the structural model of hHV1 or better describe its conformational dynamics?
- (5) Cu^{2+} , Ru^{2+} , and Ni^{2+} are presented as suitable FRET acceptors for Acd. Would Zn^{2+} also be expected to function as an acceptor in this context? If so, could structural information be derived from zinc binding independently of Trp/Tyr?
- (6) The investigated structure is most likely dimeric. Previous studies report that zinc stabilizes interactions between hHV1 monomers more strongly than in the native dimeric state. Could this provide an explanation for the observed zinc-dependent effects? Additionally, do the detergent micelles used in this study predominantly contain monomers or dimers?
- (7) hHV1 normally inserts into a phospholipid bilayer, as used in the reconstitution experiments. In contrast, detergent micelles may form monolayers rather than bilayers. Could the authors clarify the nature of the micelles used and discuss whether the protein is expected to adopt the same fold in a monolayer environment as in a bilayer?

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